Technical Debt Management Policy

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Summary. Draft proposal, not yet in force, for managing technical debt at Northstar Systems, including a 15 percent sprint capacity reservation, an impact and effort rating model, a debt register, review cadence and reporting. Approved current practice remains whatever approved engineering documents say.

SYNTHETIC DOCUMENT - Northstar Systems is a fictional company. This document exists for development, testing and demonstration only and is not real legal, employment, security or compliance advice.

1. Purpose and Scope

DRAFT NOTICE: This document is a DRAFT (version 0.9). It is not yet in force and has not been approved. Nothing in it is a binding requirement on any engineering team. The approved current practice is whatever the related approved documents say, in particular the Architecture Decision Record Process (ENG-011) and the engineering standards listed in ENG-001. Teams must not cite this draft as a reason to refuse, delay or reprioritise work, and must not report compliance or non-compliance against it. The target effective date recorded for this draft was 15 September 2026; that date has not taken effect because approval is still pending, and the date and every figure below may change before approval.

The purpose of the proposed policy is to give Northstar Systems Private Limited engineering a consistent way to identify, record, rate, prioritise and pay down technical debt so that delivery speed, reliability and security do not degrade over time. It is intended to apply to all engineering teams and all services of the Northstar Nexus Platform, Nexus Connect, Nexus Insight and internal platforms, once approved.

Until approval, teams continue to manage maintenance work through their existing backlog practices, the code review rules and quality gates in the approved engineering standards, and the vulnerability remediation deadlines in the approved security documents. This draft is circulated for comment to engineering managers, the Principal Architect, Site Reliability Engineering (SRE) and Product.

2. Definitions

- **Technical debt**: a known shortcut, outdated design, missing test, deferred upgrade or other condition in code, infrastructure or process that makes future change slower, riskier or costlier, and that the team could reasonably fix.

- **Deliberate debt**: debt knowingly taken on to meet a deadline, ideally recorded in an ADR (see ENG-011).
- **Accidental debt**: debt discovered later, for example a design that did not scale as expected.
- **Debt item**: one entry in the debt register describing a specific condition and a proposed remedy.
- **Impact**: the proposed 1 to 5 score of how much a debt item harms reliability, security, delivery speed or customers.
- **Effort**: the proposed 1 to 5 score of how much work is needed to fix the item.
- **Capacity reservation**: the share of each sprint's planned engineering capacity set aside for debt items.
- **Debt register**: the list of debt items for a team, kept in NorthDesk with owners and due dates.

3. Proposed Roles and Responsibilities

- **Team lead**: keeps the team's debt register current, runs the monthly debt review and ensures the capacity reservation is used.
- **Engineering manager**: approves the debt priorities for their teams and reports debt trends to the Director.
- **Product manager**: agrees the capacity reservation in sprint planning and does not reassign it to feature work except under the exceptions in section 8.
- **Principal Architect**: advises on architectural debt and links debt items to ADRs.
- **SRE**: raises reliability-related debt discovered in incident reviews and supports measurement.
- **VP Engineering**: owns this policy, reviews the quarterly debt report and approves exceptions beyond one sprint.

4. Proposed Capacity Reservation

The central proposal is that 15% of sprint capacity be reserved for technical debt. In this draft, capacity is measured in planned story points or engineer-days, whichever the team already uses, after deducting approved leave and the on-call load. Examples of the proposed calculation are below.

Team size	Sprint length	Total capacity	15% reservation
6 engineers	10 working days	60 engineer-days	9 engineer-days
8 engineers	10 working days	80 engineer-days	12 engineer-days
10 engineers, 2 on leave for 5 days	10 working days	90 engineer-days	13.5 engineer-days

Proposed rules for the reservation:

- The reservation should be planned in every sprint and should not be carried over as a lump sum to a later sprint.
- Work counted against the reservation must come from the debt register and must have a debt item with an agreed rating.
- Work required to fix a customer-facing defect, a vulnerability inside its deadline or an incident action item is not debt work and is not counted against the 15%.
- A team that spends less than 10% of capacity on debt for three consecutive sprints should explain the reason to the engineering manager.
- A team may spend more than 15% when the backlog warrants it, with the engineering manager's agreement.

The 15% figure is a starting proposal. The draft committee has asked for data from a pilot before confirming it. Some members believe that Tier-1 services should reserve more than Tier-3 services; this is an open question listed in section 11.

5. Proposed Rating Model

5.1 Impact and effort

Debt items would be rated by impact and effort. Each item receives an impact score from 1 to 5 and an effort score from 1 to 5, assigned by the team lead with a second engineer. The following table describes the proposed impact scale.

Impact	Description
1	Minor annoyance; slows one engineer occasionally
2	Slows the team measurably; workaround is simple
3	Causes recurring defects or slows several teams
4	Threatens reliability of a Tier-1 or Tier-2 service, or blocks a planned feature
5	Risk of an outage, data loss, security exposure or breach of a contractual commitment

The proposed effort scale is: 1 is up to one engineer-day, 2 is up to one week, 3 is up to one sprint, 4 is up to one quarter and 5 is more than one quarter or requires several teams.

5.2 Priority score

The proposed priority is impact divided by effort, so that high-impact, low-effort items come first. Items with impact 5 are treated as urgent regardless of effort and must be scheduled in the next sprint after rating. Where two items have the same ratio, the item on the higher service tier goes first. The following example table shows the proposed method.

Item	Impact	Effort	Priority (impact / effort)
Replace hand-rolled retry loop in Webhook sender	4	2	2.0
Remove unused feature flags in Pipeline editor	2	1	2.0
Migrate scheduler off deprecated queue client	4	3	1.33
Rewrite connector framework	3	5	0.6

The rewrite item scores low on the ratio, so the team would be expected to explore smaller incremental steps or raise an ADR rather than commit a quarter of capacity at once.

5.3 Categories

Each item would also receive one category to support reporting: code quality, test coverage, architecture, dependencies and upgrades, infrastructure and operations, documentation, security hygiene or data and schema. Dependency debt includes unmaintained libraries, which must also be assessed under the open-source and third-party library policy (ENG-015).

6. The Debt Register

6.1 Contents

The proposal is that every team keep a debt register in NorthDesk. A register entry would contain a short title, description, service and tier, category, impact and effort scores, the date raised, the

person who raised it, an owner, a target sprint, a link to the related ADR or incident review if any, and the status. Statuses proposed are Open, Planned, In Progress, Done and Declined. Declined items must state why, for example "accepted risk, reviewed in 12 months".

6.2 Who may raise debt

Any engineer may raise a debt item. Reviewers in the code review process (see ENG-001) may ask the author to log a debt item instead of fixing something in the same pull request when the fix would make the change too large. Pull request size is recommended at no more than 400 changed lines, and a debt item is the intended way to capture the remaining work.

6.3 Linking to ADRs

When an ADR (ENG-011) is Accepted with a consciously taken shortcut, the author would be expected to log the shortcut as a deliberate debt item with a proposed repayment date. That repayment date should be no later than the ADR checkpoint date.

7. Proposed Reviews and Reporting

- **Monthly team debt review**: 30 minutes, led by the team lead, to re-rate items, close finished ones and plan the next sprint's reservation.
- **Quarterly report**: the engineering manager would send a one-page summary to the VP Engineering showing the number of open items by impact, items closed, the percentage of capacity spent and the age of the oldest high-impact items.
- **Ageing rule (proposal)**: items rated impact 4 or 5 that are older than 180 days should be reviewed by the Director, who may reprioritise or formally accept the risk.
- **Retrospective use**: SRE would flag incidents whose cause is a known debt item so that the register can show the cost of delay.

Metrics proposed for the pilot include the debt register size, the age profile, the ratio of debt closed to debt raised, change failure rate and lead time for changes. The metrics would be used to improve planning, not to rank individual engineers.

8. Proposed Exceptions

Under the proposal, the VP Engineering may approve a temporary reduction of the reservation for a single team for no more than one sprint, for example before a customer launch or during a company-declared freeze period. A second consecutive reduction would need a written plan to recover the lost capacity. A reduction would never remove the requirement to record new debt in the register. Debt associated with security vulnerabilities is not eligible for deferral through the reservation process, because vulnerability deadlines in the security documents continue to apply.

9. Worked Examples (Illustrative Only)

These examples show how the policy would work if approved. They do not describe current obligations.

Example 1: ordinary sprint planning. The Workspace Service team has 8 engineers and a 10-day sprint, so capacity is 80 engineer-days. Under the proposal, 12 engineer-days would be reserved for debt. The team lead picks two items from the register with priority 2.0 and 1.33, totalling 11 engineer-days, plus one half-day cleanup.

Example 2: customer defect. During the same sprint a P2 customer defect requires 6 engineer-days. That work is a defect fix, not debt, so it would not use the 15% reservation. The sprint plan drops a feature story instead.

Example 3: deliberate shortcut. Under ADR 0059, a team accepts a polling implementation for Webhook retry status instead of event streaming to meet a customer launch. The author logs a deliberate debt item rated impact 3, effort 3, with a repayment date of the ADR checkpoint six months later.

Example 4: old high-impact item. A debt item rated impact 5 (single point of failure in the Pipeline scheduler lock) has stayed Planned for 200 days. Under the ageing proposal the Director would review it and either schedule it immediately or record a documented risk acceptance with an expiry date.

9.1 Proposed interaction with releases and freezes

The draft suggests that debt work follow the same release rules as any other change. Production deployments would stay inside the Tuesday to Thursday window, every change would still need a NorthDesk change ticket and a rollback plan, and quarter-end freeze periods would still apply. Debt items that need risky production changes, such as a database migration, should be scheduled early in a sprint so that they can reach the Tuesday release train rather than being squeezed into the last days.

The draft also proposes that a team close a debt item only when its remedy is verified in production, for example when an alert has stopped firing or a build has become faster, and that the closing note state the evidence. Items closed without evidence would be reopened at the next monthly review. These verification rules are suggestions to be tested in the pilot, and reviewers are asked to say whether they are too heavy for small items rated effort 1.

10. Transition Plan if Approved

The proposed rollout has three steps.

1 A pilot with four teams (two Tier-1, one Tier-2, one Tier-3) for two full quarters, collecting the measures in section 7.
2 A review by the VP Engineering and the Principal Architect, who may adjust the 15% figure, the scales or the cadence.
3 A formal approval and a published effective date, after which the register becomes mandatory for all teams within one quarter.

Until the approval is announced, no team is required to take any step in this document. Teams that wish to join the pilot may ask their engineering manager.

11. Open Questions for Reviewers

- Should the reservation differ by service tier, or should all teams use the same 15%?
- Should security hygiene debt have a separate quota?
- Is impact divided by effort the right priority formula, or should risk be weighted more heavily?
- Should the register live in NorthDesk or in the repository next to the code?
- How should debt in services owned by two teams be assigned?
- How should the policy treat vendor-supplied components that Northstar cannot change?
- What is the right ageing threshold for high-impact items: 90, 180 or 365 days?

Comments are collected by the VP Engineering's office. Because this document is a draft, answers to these questions are not yet decided.

12. Exceptions and Escalation

Because the policy is not in force, there is no exception process to follow. Any question about how to treat a maintenance item today should go to the engineering manager, who applies the approved documents. Disagreements about priority between maintenance work and features are resolved through the normal escalation route to the Director and then the VP Engineering. Security vulnerabilities must be handled under the approved vulnerability SLAs, and suspected security issues reported to the Security Desk.

13. Frequently Asked Questions

Is the 15% reservation mandatory today? No. The 15% reservation is only a proposal in this draft and is not in force.

Which document governs technical debt today? Approved practice is whatever the approved engineering documents say, including ENG-011 for architecture decisions and the standards in ENG-001. No approved document currently requires a fixed share of sprint capacity for debt.

Can a manager refuse a debt item because the policy is not approved? Managers prioritise the backlog as they normally do. The draft gives them no obligation either way.

When will it take effect? The target date was 15 September 2026, but no approval has been given, so there is no effective date at present.

14. Related Documents

- ENG-011 is the approved Architecture Decision Record Process, under which deliberate shortcuts are recorded.
- ENG-001 sets the approved engineering standards for code review, testing and delivery that apply now.
- ENG-015 governs dependencies and open-source licences, which feed dependency debt.

15. Revision History

Version	Date	Change
0.5	2026-05-18	First internal draft circulated to engineering managers
0.9	2026-09-15	Added rating model, register and pilot plan; still pending approval